

IT Ops Console

A serverless IT-operations tool built as a portfolio project for Clockwork's IT Operations Specialist posting (Minneapolis, macOS-based environment).

Live demo: <https://d198vhce9vcowy.cloudfront.net>

Source code: <https://github.com/bcole84/clockwork-it-ops-console>

Why this exists

Rather than just list skills on a resume, I built a small working tool covering three of the role's core duties, plus the documentation habit the posting explicitly calls for. Everything here runs on my own AWS account and was scoped to stay inside AWS's **always-free** tier.

What it does

Feature	Job posting duty it demonstrates
Asset Tracker	"Manage IT assets... procure, configure, track, maintain, retire... using MDM"
Onboarding & Offboarding	"Manage employee onboarding and offboarding"
Helpdesk Tickets	"Triage requests, troubleshoot problems, resolve support tickets"
Knowledge Base (5 playbooks)	"Document and share knowledge... playbooks, training materials"

Asset Tracker

Tracks laptops and other hardware: type, serial number, assigned employee, status (In Stock / In Use / In Repair / Retired), purchase date, and notes.



IT Ops Console
Asset tracking · Onboarding & offboarding · Helpdesk tickets · Knowledge base

API connected

Assets

Onboarding & Offboarding

Tickets

Knowledge Base

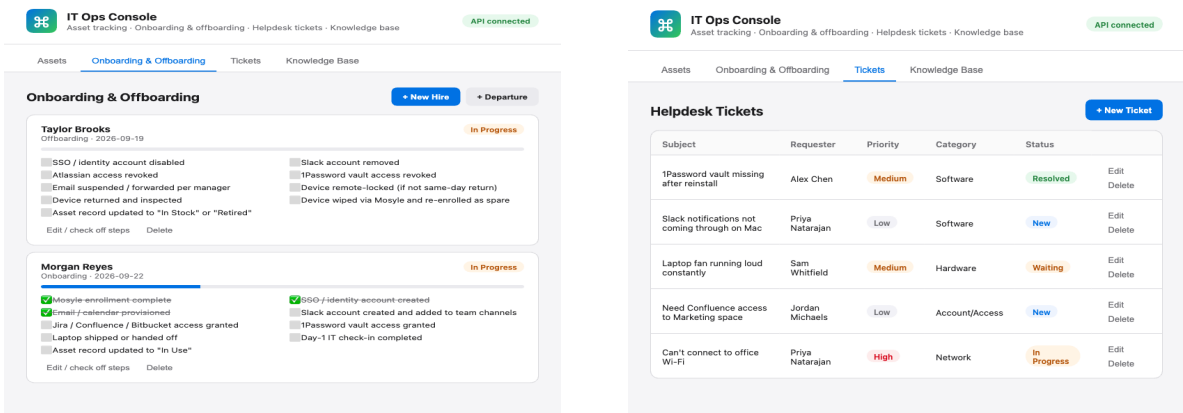
IT Assets

+ New Asset

Type	Serial	Assigned To	Status	Purchased	Notes	
MacBook Pro 16" M2 Max	FVFXC2AB7890	Sam Whitfield	Retired	2021-09-30	End of 4-year refresh cycle; pending wipe and recycling	Edit Delete
MacBook Air 13" M2	FVFXC2AB3456	—	In Repair	2024-08-15	Battery swelling - sent to Apple Store for service	Edit Delete
MacBook Pro 14" M3 Pro	FVFXC2AB5678	Jordan Michaels	In Use	2025-05-22	Engineering build; 36GB RAM	Edit Delete
MacBook Air 13" M2	FVFXC2AB9012	—	In Stock	2024-11-01	Spare pool - ready to enroll in Mosyle	Edit Delete
MacBook Air 15" M3	FVFXC2AB1234	Priya Natarajan	In Use	2025-02-10	AppleCare+ through 2028-02	Edit Delete

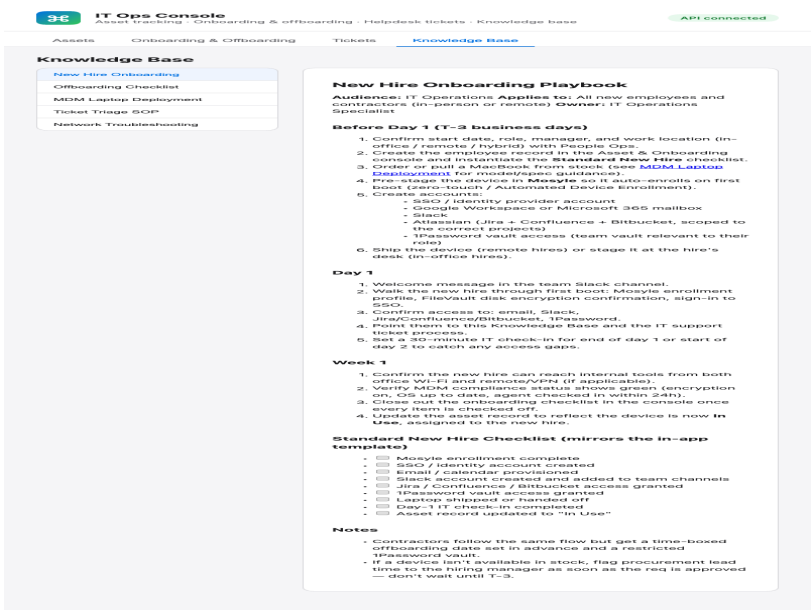
Onboarding & Offboarding, Tickets, Knowledge Base

Onboarding/offboarding checklists track per-employee progress against a standard template (Mosyle enrollment, SSO, Slack, Atlassian, 1Password, laptop handoff). Tickets cover intake and triage with priority/category/status. The Knowledge Base renders five macOS/Mosyle-specific playbooks I wrote directly from the app.



Onboarding & Offboarding

Helpdesk Tickets

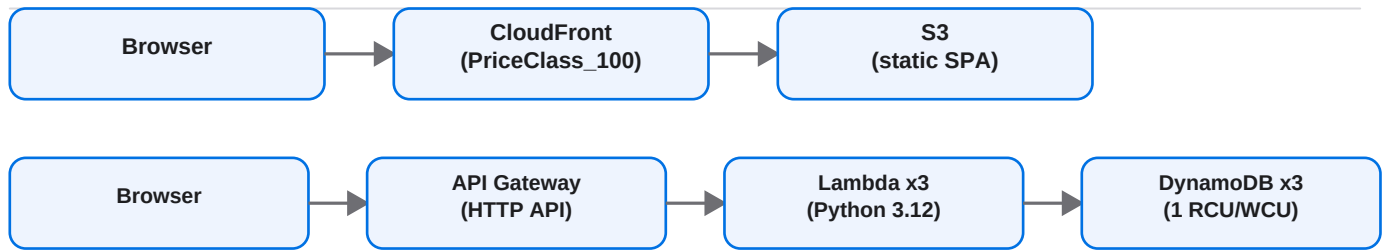


Knowledge Base

Five playbooks, written the way I'd actually hand them to a new IT hire on day one:

- New Hire Onboarding
- Offboarding Checklist
- MDM Laptop Deployment (Mosyle)
- Ticket Triage SOP
- Network Troubleshooting (macOS)

Architecture



Fully serverless, no fixed-hourly-cost resources (no NAT Gateway, VPC, RDS, or ALB). Infrastructure is defined as a single CloudFormation template and deployed via the AWS CLI (`cloudformation package / deploy`) rather than CDK/SAM, since the build machine had no Node.js toolchain installed.

Technical decisions worth noting

- **DynamoDB at provisioned 1 RCU / 1 WCU per table** (3/3 total) — inside AWS's *always-free* allowance (25/25), not just the 12-month one.
- **API Gateway HTTP API instead of REST API** — ~70% cheaper per request beyond free tier, simpler built-in CORS.
- **CloudFront PriceClass_100** — cheapest edge-location tier, sufficient for a demo.
- **Least-privilege IAM** — each Lambda's execution role is scoped to only its own DynamoDB table.
- Two real bugs found and fixed during deployment: HTTP API's automatic CORS doesn't intercept `OPTIONS` when a route uses `ANY` (fixed with an early return in each Lambda); DynamoDB's `Decimal` type isn't JSON-serializable by default (fixed with a custom JSON encoder).

What I'd add next

Kept out of scope on purpose, to keep this honest about what it is: Cognito-based auth instead of a shared-secret header; a dedicated least-privilege IAM deploy user instead of root credentials; CI/CD instead of a manual deploy script.